

$$\text{binomial therom: } (x+a)^r = \sum_{b=0}^r \left[\binom{r}{b} x^b a^{r-b} \right]$$

let $a = 1$, 1^n always $=1$ so all x terms are just multiplied by 1

$$(x+1)^r = \sum_{b=0}^r \left[\binom{r}{b} x^b 1 \right]$$

let $x' = x - 1$ to cancel the $+1$ and produce x^r

$$((x-1)+1)^r = x^r = \sum_{b=0}^r \left[\binom{r}{b} (x-1)^b \right]$$

bring out the $(x-1)^r$ to get recursssion (by applying the identity to that term)

$$x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-1)^b \right] + (x-1)^r$$

$$x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-1)^b \right] + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-2)^b \right] + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-3)^b \right] + (x-3)^r$$

$$x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-1)^b \right] + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-2)^b \right] + \dots + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-x)^b \right] + (x-x)^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} b^c \right] \right] + 0^r$$

apply the identity to the b^c terms assymetrically to get sums of x^0

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} b^c \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{0} b^0 + \binom{r}{1} b^1 + \binom{r}{2} b^2 + \dots + \binom{r}{r-1} b^{r-1} \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\binom{r}{0} b^0 + \binom{r}{1} \sum_{c=0}^{b-1} \left[\binom{1}{0} c^0 \right] + \binom{r}{2} \sum_{c=0}^{b-1} \left[\binom{2}{0} c^0 + \binom{2}{1} \sum_{d=0}^{c-1} \left[\binom{1}{0} d^0 \right] \right] + \dots + \binom{r}{r-1} b^{r-1} \right] + 0^r$$

$$x^r = \binom{r}{0} \sum_{b=0}^{x-1} [b^0] + \binom{r}{1} \binom{1}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + \binom{r}{2} \binom{2}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + \binom{r}{2} \binom{2}{1} \binom{1}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [d^0] + \dots + \binom{r}{r-1} b^{r-1} + 0^r$$

$$x^r = \binom{r}{0} \sum_{b=0}^{x-1} [b^0] + \left(\binom{r}{1} \binom{1}{0} + \binom{r}{2} \binom{2}{0} \right) \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + \binom{r}{2} \binom{2}{1} \binom{1}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [d^0] + \dots + \binom{r}{r-1} b^{r-1} + 0^r$$

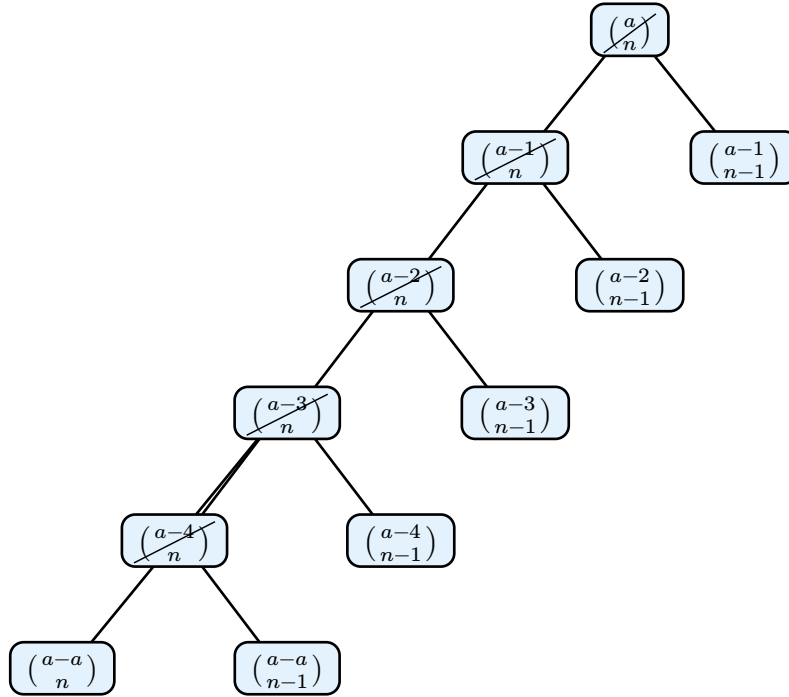
let the product and sum of these outer binom terms grouped by sum level (n) , $= D_{r,n}$

$$x^r = D_{r,1} \sum_{b=0}^{x-1} [b^0] + D_{r,2} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + D_{r,3} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [d^0] + \dots + D_{r,r} \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} \dots}^{r \text{ sums}} \sum_{n_n=0}^{n_{n-1}-1} [n_n^0] + 0^r$$

pascals identity

$$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$$

apply pascals identity recursively to lhs



$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\binom{b}{n-1} \right] + \binom{0}{n}$$

apply the identity to the inner $\binom{b}{n-1}$

$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\binom{c}{n-2} \right] + \binom{0}{n-1} \right] + \binom{0}{n}$$

expanding the inner term n-1 times, for n total sums, causes its inner term to be 1, as $\binom{a \geq 0}{0} = 1$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} \left[\binom{n_n}{n-n} \right] + \binom{0}{n-(n-1)} \dots \right] \binom{0}{n-2} \right] + \binom{0}{n-1} \right]}_{n \text{ sums}} + \binom{0}{n}$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} \left[\binom{n_n}{0} \right] + \binom{0}{n-(n-1)} \dots \right] \binom{0}{n-2} \right] + \binom{0}{n-1} \right]}_{n \text{ sums}} + \binom{0}{n}$$

as we only expanded n-1 times the $\binom{0}{n_{n-1}}$ only reach $\binom{0}{n-(n-1)} = \binom{0}{1} = 0$, as $\binom{0}{n > 0} = 0$

if $n = 0$, the sums arent expanded and only the $\binom{0}{n}$ remains

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} [1] \right] \right] \right]}_{n \text{ sums}} + \binom{0}{n}$$

the sums in the binomial theorem expansion and pascals identity expansions have the same bounds change

$$x^r = D_{r,1} \sum_{b=0}^{x-1} [1] + D_{r,2} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [1] + D_{r,3} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [1] + \dots + D_{r,r} \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} \dots}^{r \text{ sums}} \sum_{n_n=0}^{n_{n-1}-1} [1] + 0^r$$

$$x^r = D_{r,1} \binom{x}{1} + D_{r,2} \binom{x}{2} + D_{r,3} \binom{x}{3} + \dots + D_{r,r} \overbrace{\binom{x}{r}}^{r \text{ sums}} + 0^r$$

$$x^r = \sum_{b=1}^r \left[D_{r,b} \binom{x}{b} \right] + 0^r$$

$$\text{let } D_{r,0} = 0^r$$

$$x^r = \sum_{b=0}^r \left[D_{r,b} \binom{x}{b} \right]$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} b^c \right] \right] + 0^r$$

every expansion of b^c adds a sum layer to the expanded terms, the $\sum_{b=0}^{x-1}$ is this accumulator

each new layer of b_n^c adds a sum to the expanded terms, so each new layer can only produce differences upto n (because it has a term with a minimum of n sums)

therefore to get the n th difference you would need to expand n times, ($n-1$ times if you count the base identity as the 0th expansion because it adds a outer sum to the terms); deliberate expansion of terms with r

$$x^r = \sum_{b=0}^{x-1} \left[\binom{r}{0} b^0 + \binom{r}{1} b^1 + \binom{r}{2} b^2 \dots \binom{r}{r-1} b^{r-1} \right] + 0^r$$

$$b^0 = 1$$

$$b^1 = \sum_{c=0}^{x-1} \left[\overbrace{\binom{1}{0} c^0}^1 \right]$$

$$b^2 = \sum_{c=0}^{x-1} \left[\overbrace{\binom{2}{0} c^0}^1 + \binom{2}{1} \sum_{d=0}^{c-1} \left[\overbrace{\binom{1}{0} d^0}^1 \right] \right]$$

each layer contains a 1 term it will have L (layer) amount of sums (for $r \geq 0$)

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=1}^{r-1} \left[\binom{r}{c} b^c \right] + \overbrace{\binom{r}{0} b^0}^{1 \text{ at } r \geq 1} \right] + \widehat{0^r}^{1 \text{ at } r=0}$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} b^c \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^r \left[D_{r,b} \binom{x}{b} \right]$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} \sum_{d=0}^c \left[D_{c,d} \binom{b}{d} \right] \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\binom{r}{0} \sum_{d=0}^0 \left[D_{0,d} \binom{b}{d} \right] + \binom{r}{1} \sum_{d=0}^1 \left[D_{1,d} \binom{b}{d} \right] + \dots + \binom{r}{r-1} \sum_{d=0}^{r-1} \left[D_{r-1,d} \binom{b}{d} \right] \right] + 0^r$$

grouping the terms by $\binom{b}{d}$ tells the coefficients of each difference level, this will produce a formula for $D_{r,c}$ in terms of $D_{r,c}$ from lower degree terms.

since every time you lower the degree you loose a difference level until D_0 , there will be a falling number of each difference level in the sum. $\sum^{r-1} D_0 + \sum^{r-2} D_1 + \dots \sum^{r-r} D_{r-1}$ (grouping the inner sum, excluding $\sum^{x-1}$)

$$x^r = \sum_{b=0}^{x-1} \left[\binom{r}{0} \sum_{d=0}^0 \left[D_{0,d} \binom{b}{d} \right] + \binom{r}{1} \sum_{d=0}^1 \left[D_{1,d} \binom{b}{d} \right] + \dots + \binom{r}{r-1} \sum_{d=0}^{r-1} \left[D_{r-1,d} \binom{b}{d} \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\overbrace{\left[\binom{r}{0} D_{0,0} + \binom{r}{1} D_{1,0} + \dots + \binom{r}{r-1} D_{r-1,0} \right]}^{D_{r,0+1}} \binom{b}{0} + \dots + \overbrace{\left[\binom{r}{r-1} D_{r-1,r-1} \right]}^{D_{r,r-1+1}} \binom{b}{r-1} \right] + 0^r$$

because of outer sum the difference level has 1 added to it, the grouped coefficients on $\binom{b}{n}$ tells the difference level of the n th difference as this would expand to n falling sums

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} [D_{c,0} \binom{r}{c}] \binom{b}{0} + \sum_{c=1}^{r-1} [D_{c,1} \binom{r}{c}] \binom{b}{1} + \dots + \sum_{c=r-1}^{r-1} [D_{c,r-1} \binom{r}{c}] \binom{b}{r-1} \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{b}{c} \sum_{d=c}^{r-1} [D_{d,c} \binom{r}{d}] \right] \right] + 0^r$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} [1] \right] \right] \right]}_{n \text{ sums}} + \binom{0}{n}$$

$$x^r = \sum_{c=0}^{r-1} \left[\overbrace{\binom{x}{c+1}}^{\text{Difference sums (c+1)}} \overbrace{\sum_{d=c}^{r-1} [D_{d,c} \binom{r}{d}]}^{\text{Difference coefficients}} \right] + \overbrace{\widehat{0^r}}^{\text{0th Difference}}$$

$$D_{r,0} = 0^r$$

$$D_{0,0} = 1$$

$$D_{r,c+1} = \overbrace{\sum_{d=c}^{r-1} [D_{d,c} \binom{r}{d}]}^{\text{Difference coefficients}}$$

$$D_{r,c} = \overbrace{\sum_{d=c-1}^{r-1} [D_{d,c-1} \binom{r}{d}]}^{\text{Difference coefficients}}$$

$$D_{r,c} = \sum_{d=c-1}^{r-1} [D_{d,c-1} \binom{r}{d}]$$

$$D_{r,c} = \sum_{d=c-1}^{r-1} \left[\sum_{e=c-2}^{d-1} [D_{e,c-2} \binom{d}{e}] \binom{r}{d} \right]$$

expand c-1 times, (c times if you count the base differences identity as the 0th expansion)

$$D_{r,c} = \overbrace{\sum_{d=c-1}^{r-1} \left[\sum_{e=c-2}^{d-1} \left[\sum_{f=c-3}^{e-1} \left[\dots \sum_{n_{c-1}=c-c}^{n_{c-2}-1} [D_{n_{c-1},c-c} \binom{n_{c-2}}{n_{c-1}}] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

$$D_{r,c} = \sum_{d=c-1}^{r-1} \left[D_{d,c-1} \binom{r}{d} \right]$$

since x^r only has differences upto level r , $D_{r,(c>r)}$ can be treated as 0

$$\begin{aligned} D_{r,c} &= \sum_{d=0}^{r-1} \left[D_{d,c-1} \binom{r}{d} \right] \\ D_{r,c} &= \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-1}=0}^{n_{c-2}-1} \left[D_{n_{c-1},c-c} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right] \\ &\quad \text{c sums} \\ D_{r,c} &= \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-1}=0}^{n_{c-2}-1} \left[D_{n_{c-1},0} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right] \\ &\quad \text{c sums} \\ D_{r,0} &= 0^r \end{aligned}$$

$$D_{r,c} = \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-1}=0}^{n_{c-2}-1} \left[0^{n_{c-1}} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right] \quad \text{c sums}$$

since $0^r = 1$ only at $r = 0$, the inner sums $0^{n_{c-1}}$ will only be 1 at the start, since $0^{\neq 0}$ is 0 the $\binom{n_{c-2}}{n_{c-1}}$ will cancel out when not at the start of the sum, the sum only produces a single non zero run at its start, as long as the inner most upper bound is not negative, if $r \geq c$ the inner sum is guaranteed to run at least once

the inner sum only runs once if upper bound is not negative

$$\begin{aligned} D_{r,c} &= \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-2}=1}^{n_{c-2}-1} [1] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right] \\ &\quad \text{c sums} \\ D_{r,c} &= \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-2}=0}^{n_{c-3}-1} \left[\sum_{n_{c-2}=1}^{n_{c-2}-1} [1] \binom{n_{c-3}}{n_{c-2}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right] \\ &\quad \text{c sums} \end{aligned}$$

for the inner most sum to run the outer sums lower bound must be at least 1, $n_{c-2} \geq 1$, if it is 0 it will not run

$$1 - 0^{n_{c-2}} \text{ is only 1 if } n_{c-2} \neq 0 \text{ or } n_{c-2} > 0 \text{ and } n_{c-2} < 0$$

since n_{c-2} starts at 0 and goes upto $n_{c-3} - 1$ it can be negative or its parent sum wouldn't have run it

$$\begin{aligned} D_{r,c} &= \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-2}=0}^{n_{c-3}-1} \left[\overbrace{[1 - 0^{n_{c-2}}] \binom{n_{c-3}}{n_{c-2}} \dots \binom{e}{f}}^{c-1 \text{ sums}} \right] \binom{d}{e} \right] \binom{r}{d} \right] \right] \\ &\quad \text{c-1 sums} \\ &+ \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\sum_{n_{c-2}=0}^{n_{c-3}-1} \left[\overbrace{[-0^{n_{c-2}}] \binom{n_{c-3}}{n_{c-2}} \dots \binom{e}{f}}^{c-1 \text{ sums}} \right] \binom{d}{e} \right] \binom{r}{d} \right] \right] \\ &\quad \text{c-1 sums} \end{aligned}$$